# Theoretical and Numerical Analysis of Measurement Limits in Quantum Image Representations: Qubit Lattice, FRQI, and NEQR

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***Abstract*—With the advancement of quantum computing, quantum image representations (QIRs) have been widely studied as platforms for quantum image processing. While previous studies have focused on theoretical qubit counts and circuit complexities, there has been limited quantitative evaluation of image reconstruction performance from quantum states. In this study, we clarify the theoretical upper bounds of restoration performance for three representative QIRs: qubit lattice (QL), flexible representation of quantum images (FRQI), and novel enhanced quantum representation (NEQR). Also, we validate them through large-scale GPU-based simulations using cuQuantum. Specifically, we (1) derive a prediction formula for PSNR in probabilistic schemes like QL and FRQI, independent of image content and size; (2) show that deterministic schemes like NEQR require a number of samples following a harmonic number for complete restoration; and (3) perform large-scale simulations to verify these findings. These results provide fundamental insights for designing QIRs and their applications on NISQ devices.**

***Index Terms*—quantum image representation, quantum image processing, cuQuantum, FRQI, NEQR**

## I. Introduction

Image processing on quantum computers [1] has attracted attention and there are various survey papers [2]–[12]. Using the quantum superposition [13], quantum computers have a different level of parallelism than classical computers [14], [15]. While classical parallel computing uses SIMD, multi-core threading [16], tiling [17], pipeline burst cache pipeline processing [18], and Tensor Cores [19], quantum computing uses quantum bits for parallelization. This quantum parallelism is significantly higher than the classical one; thus, quantum computers are expected to enable high-speed computing.

Quantum image processing is based on quantum image representation (QIR) [10]. There are various QIRs, such as qubit lattice (QL) [20], entangled image [21], real ket [22], flexible representation of quantum images (FRQI) [23], and quantum probability image encoding (QPIE) [24]. These encode intensity probabilistically, while novel enhanced quantum representation (NEQR) [25] and bitplane representation of quantum images (BRQI) [26] encode it deterministically.

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Each QIR has been theoretically evaluated in terms of the required number of qubits, computational complexity, and the number of quantum gates [12]. Also, quantum pixel representation (QPIXL) [27] and QCrank and QBArt [28] techniques can reduce the depth of the quantum circuit of various QIRs, including FRQI and NEQR, for the noisy intermediate-scale quantum (NISQ) era. In addition, various image processing tasks achievable with these QIRs have been demonstrated [6], [12], [29]. Recent studies have even demonstrated implementations on actual quantum hardware for representative QIRs, such as FRQI and NEQR [30], [31]. Most prior studies on quantum image processing have focused on expanding QIR structures, analyzing theoretical qubit counts, circuit complexities, and demonstrating possible image processing tasks. However, quantitative evaluation of image reconstruction performance from quantum states—especially regarding the number of samples required to achieve sufficient image quality—has been limited. Existing works often present examples after a fixed number of measurements or use quantum-state-only metrics, such as the quantum-based image fidelity metric (QIFM) [32], which do not assess the quality of reconstructed images.

This paper clarifies the theoretical upper bounds of restoration performance for three representative QIRs (QL, FRQI, and NEQR) and validates them via large-scale simulation using cuQuantum [33]. This paper shows that the theoretical upper limit of the QIR provides an estimate of the required sample size. Our contributions are summarized as follows:

- Demonstrate that for probabilistic schemes of QL and FRQI, the PSNR of reconstructed images can be predicted solely from the number of samples per pixel, independent of image content and size.
- Show that deterministic schemes of NEQR require a number of samples following a harmonic number to achieve full restoration.
- Leverage cuQuantum acceleration to conduct large-scale simulations, enabling validation of theoretical predictions beyond the conventional quantum simulators [34].

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## II. Quantum Image Representation

### A. Qunatum States

The fundamental unit of a quantum computer is a qubit. A bit of a classical computer can be a digital value of either 0 or 1, whereas a qubit can be a probabilistic value of either 0 or 1, which is in a superposition state [13]. The quantum state $|\psi\rangle$ of a qubit is expressed by using a linear combination of the ground state $|0\rangle$ and the excited state $|1\rangle$.

$$|\psi\rangle = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \alpha |0\rangle + \beta |1\rangle , \tag{1}$$

where, $\alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1$. The quantum bit $|\psi\rangle$ represents the superposition of $|0\rangle$ and $|1\rangle$ using two complex numbers $\alpha$ and $\beta$, called the probability amplitude. A qubit can store a probabilistic value, but measurements converge to either 0 or 1. In this case, the probability of zero is $|\alpha|^2$, and that of one is $|\beta|^2$. For example, if a qubit holds a probability of 0.6, measuring 10 times will likely result in 6 ones and 4 zeros being sampled, decoding 0.6. Note that higher precision reconstruction increases the number of sampling trials.

Using multiple qubits, the quantum state becomes a superposition state of combinations of more basis states. For 2 qubits, the quantum state of $|\Psi\rangle$ is represented by

$$|\Psi\rangle = \alpha |00\rangle + \beta |01\rangle + \gamma |10\rangle + \delta |11\rangle , \tag{2}$$

where $\gamma, \delta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$. Here, $|00\rangle$ indicates a state where two quantum bits are 0. In the $n$-qubits case, the quantum state $|\Psi\rangle$ is a linear combination of the basis states:

$$|\Psi\rangle = \sum_{i=1}^{2^n} c_i |i\rangle , \quad \sum_{i=1}^{2^n} |c_i|^2 = 1, c_i \in \mathbb{C}. \tag{3}$$

Here, $|i\rangle , i = i_1 i_2 \cdots i_n$, represents all $2^n$ possible basis states of $n$ bits, e.g., $|000\rangle , |001\rangle , \ldots , |111\rangle$ for $n = 3$ case. It is also possible to express an arbitrary quantum state as a superposition of linear combinations. Let $\phi_1, \phi_2, \ldots, \phi_n$ be arbitrary $n$-qubits quantum state, which is expressed by (3). The superposition is defined as follows

$$|\Psi\rangle = \sum_{i=1}^{2^n} c_i |\phi_i\rangle = \sum_{i=1}^{2^n} d_i |i\rangle , \quad \sum_{i=1}^{2^n} |d_i|^2 = 1, d_i \in \mathbb{C}. \tag{4}$$

In this case, the normalization condition requires normalization of $d_i$ in the expanded expression as the basic state, rather than normalization of $c_i$.

When multiple qubit sequences expressing a single qubit one by one, the tensor product is used, e.g., $|00\rangle = |0\rangle \otimes |0\rangle$. This expression can describe the quantum state as a combination of subsets of the quantum state using small quantum bit sequences rather than a combination of basic states.

$$\begin{aligned} |\Psi\rangle &= |\psi_1\rangle \otimes |\psi_2\rangle \otimes |\psi_n\rangle = |\psi_1 \psi_2 \cdots \psi_n\rangle , \\ &= \sum_{i=1}^{2^n} u_{i_1} \cdot u_{i_1} \cdots u_{i_n} |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_n\rangle \end{aligned} \tag{5}$$

where $|\psi_n\rangle$ is a arbitrary quantum state using one qubit of (1). This shows that the coefficients $c_i$ for each basic state of multiple qubits in (3) can be factored so that they correspond to the coefficients $u_{i_n}$ of each individual quantum bit, i.e., $c_i = u_{i_1} \cdot u_{i_1} \cdots u_{i_n}$. When $|\Psi\rangle \neq |\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle$, it is called an entangled superposition state, in which not all quantum bits can be combined by $\otimes$. When expressed by arbitrary coefficients $c_i$, it is usually an entangled state.

### B. Qubit Lattice (QL)

QL [20] is an early QIR proposed in 2003. It does not use the quantum property of superposition, but only that of probabilistic measurement of qubits, i.e., no quantum speed-up. QL has each pixel value corresponding to the probability amplitude of one qubit for each pixel. The pixel value of QL at $i = (x, y) \in S$, $|v_i\rangle$, where $S$ is a set of pixel positions and $|S| = 2^n$, is expressed as follows.

$$\begin{aligned} |v_i\rangle &= \cos\theta_i |0\rangle + \sin\theta_i |1\rangle , \quad 0 \leq \theta_i \leq \pi/2, \\ |I\rangle &= |v_1 v_2 \cdots v_n\rangle = |v_1\rangle \otimes |v_2\rangle \otimes \cdots \otimes |v_{2^n}\rangle , \end{aligned} \tag{6}$$

where $|I\rangle$ is an image represented by the qubit lattice. The qubit lattice simply represents bits of a pixel (e.g., 8-bit integer or 32-bit float) with a quantum bit and does not have parallelism in spatial pixel position and an entanglement state. The qubit lattice has a qubit per pixel.

### C. Flexible Representation of Quantum Images (FRQI)

The FRQI's pixel values are also expressed using one qubit as $|v_i\rangle = \cos\theta_i |0\rangle + \sin\theta_i |1\rangle$. In addition, a pixel in FRQI has its corresponding pixel position expressed by the additional qubit sequence $|i\rangle$, where $i = i_1 i_2 \cdots i_n$, $i_n \in \{0, 1\}^n$ [23]. FRQI contains intensity and position information for each pixel, but does not assign qubits to each pixel, unlike QL. FRQI utilizes quantum superposition to convey color and spatial information, where intensity and position are entangled. The sequence of color and spatial qubits, $|\nu_i\rangle$, is generated by the tensor product of each qubit and is formulated as follows.

$$\begin{aligned} |\nu_i\rangle &= |v_i\rangle \otimes |i\rangle , \\ |I\rangle &= \frac{1}{2^n} \sum_{i \in S} |\nu_i\rangle \otimes |i\rangle . \end{aligned} \tag{7}$$

A classical computer stores $n$ binary values in $n$ bits, while a quantum computer can represent complex amplitudes $n$ in simple cases. Due to quantum superposition, it can encode $2^n$ possible configurations simultaneously. FRQI allows parallel calculation of all superimposed pixels using parallel quantum circuits. On the other hand, since only the binary value of one pixel is observed, the number of samples increases.

### D. Novel Enhanced Quantum Representation (NEQR)

NEQR extends FRQI as a digital intensity representation proposed in 2013 [25]. Like FRQI, NEQR utilizes pixel values and positions with superposition. However, it differs from FRQI in that it uses multiple quantum bits to represent pixel values as digital. When the maximum pixel value is represented by $2^d$ with the representation of $d$ bits as $f = C_1 C_2 \ldots C_d \in \{0, 1\}^d$, NEQR is represented as follows.

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$$|I\rangle = \frac{1}{2^n}\sum_{i\in S}|C_{i_1}\rangle\otimes\cdots\otimes|C_{i_d}\rangle\otimes|i\rangle = \frac{1}{2^n}\sum_{i\in S}|f_i\rangle\otimes|i\rangle \quad (8)$$

NEQR also has entangled pixel values and positions, and has high parallelability. In addition, because intensity is digital information quantized to 8 bits, sampling requires only digital restoration of 0 or 1, which significantly reduces the number of measurements.

## III. Theoretical Sampling Performance

### A. Quantum Sampling

Quantum computers operate quantum states by quantum gates, and the resulting states are obtained to classical computer through probabilistic quantum measurements. In usual destructive measurements (projection measurements), the quantum states are lost because they converge to binary states. Therefore, it is necessary to obtain the quantum state by repeating measurements many times. With FRQI and NEQR, a single measurement provides only the probabilistic contribution to the pixel value of a single pixel; thus, multiple samplings are required to recover the image. We achieve this using cuQuantum's measurement (*custatevecSamplerSample*).

### B. Theoretical formula for Qubit Lattice and FRQI

Let an image be $\boldsymbol{I} = \{p_1, p_2, \ldots, p_N\} \in \mathbb{R}^N$ with pixel count $N = W \times H$ and each pixel value $p_i$ is a non-negative real number in the range $[0.0, 255.0]$. The image restoration problem using FRQI is formulated as generating a reconstructed image by performing $M$ random samples, treating the function encoding $\boldsymbol{I}$ as a probability density function. For simplicity, the image is treated as a one-dimensional vector by flattening $(x, y)$ into a single index $i$, without loss of generality.

In FRQI, the probability $q_i$ of measuring $|1\rangle$ in the color qubit for a value $p_i$ at a position $i$ is defined as:

$$q_i = \sin^2\theta_i = \sin^2\left(\frac{\pi p_i}{2\cdot 255}\right) \quad (9)$$

Let the number of samples at $i$ be $m_i$ and that of measurements $|1\rangle$ be $o_i$. The reconstruction probability $\hat{q}_i$ is defined as:

$$\hat{q}_i = \frac{o_i}{m_i}. \quad (10)$$

Since each pixel is sampled as a uniform distribution, the number of samples at each pixel position $m_i$ is expected to be the following constant as the average number in the limit.

$$m = \frac{M}{N}. \quad (11)$$

The restored pixel intensity $\hat{p}_i$ from the restored probability $\hat{q}_i = o_i/m$ is defined as follows:

$$\hat{p}_i = \frac{2\cdot 255}{\pi}\arcsin\left(\sqrt{\frac{o_i}{m}}\right). \quad (12)$$

This maps the values from 0 to 255 so that they appear as a function of $\sin^2$ instead of a uniform distribution. This quantum measurement is a repetition of Bernoulli trials with a success probability $q_i$. In $m$ independent samplings, the probability that each selects the $i$-th pixel is $\hat{q}_i \sim \mathrm{Binomial}(m, q_i)/m$, and the number of hits for a pixel $i$, $o_i$, follows a binomial distribution. The expected value of the residual per pixel between the ideal and reconstructed images is expressed as follows:

$$\mathbb{E}\left[(\hat{p}_i - p_i)^2\right] = \sum_{k=1}^{m}\binom{m}{k}q_i^k(1-q_i)^{m-k}\left(\frac{2\cdot 255}{\pi}\cdot\arcsin\left(\sqrt{\frac{k}{m}}\right) - p_i\right)^2. \quad (13)$$

The cost of computing (13) is exactly high since it involves the summation and combination of a large $m$; thus, the calculation of an exact solution is limited to small $m$.

We solve this computational problem using the delta method in statistics. Let the inverse function be $p_i = f(q_i)$, then

$$\mathbb{E}\left[(\hat{p}_i - p_i)^2\right] = \mathbb{E}\left[(f(\hat{q}_i) - f(q_i))^2\right]. \quad (14)$$

When we do a Taylor expansion of the function $f(\hat{q}_i)$ around $q_i$, we obtain the following form:

$$f(\hat{q}_i) = f(q_i) + f'(q_i)(\hat{q}_i - q_i) + R_2. \quad (15)$$

Here, $R_2$ is the remainder term from the second order onward. $f$ is $\arcsin$ function; thus, its derivative is as follows:

$$f'(q_i) = \frac{255}{\pi\sqrt{q_i(1-q_i)}}. \quad (16)$$

Taking the square error, we can obtain the expected value:

$$\begin{aligned}(f(\hat{q}_i) - f(q_i))^2 &= f'(q_i)^2(\hat{q}_i - q_i)^2 + 2f'(q_i)(\hat{q}_i - q_i)R_2 + R_2^2,\\ \mathbb{E}\left[(f(\hat{q}_i) - f(q_i))^2\right] &= f'(q_i)^2\cdot\mathbb{E}\left[(\hat{q}_i - q_i)^2\right]\\ &\quad + 2f'(q_i)\cdot\mathbb{E}\left[(\hat{q}_i - q_i)R_2\right] + \mathbb{E}\left[R_2^2\right]. \quad (17)\end{aligned}$$

Since $q_i$ is the mean value of $\hat{q}_i$ in the limit, the expectation of $(\hat{q}_i - q_i)^2$ is the variance. In addition, the variance of the binomial distribution is given by

$$\mathbb{E}[(\hat{q}_i - q_i)^2] = \mathrm{Var}[\hat{q}_i] = \frac{q_i(1-q_i)}{m}. \quad (18)$$

As the number of iterations increases, approaching $\hat{q} \to q$, the remainder term converges to 0 and can be ignored in the formula. Consequently, the formula is approximated as:

$$\begin{aligned}\mathbb{E}\left[(\hat{p}_i - p_i)^2\right] &\approx f'(q_i)^2\cdot\mathrm{Var}[\hat{q}_i]\\ &= \frac{255^2}{\pi^2 q_i(1-q_i)}\cdot\frac{q_i(1-q_i)}{m} = \frac{255^2}{m\pi^2}. \quad (19)\end{aligned}$$

The entire image's MSE measured $m$ times per pixel is

$$\mathrm{MSE}_m = \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}\left[(\hat{p}_i - p_i)^2\right] \approx \frac{255^2}{m\pi^2}. \quad (20)$$

Note that since the same average count $m$ is assumed for each pixel, the value is the same as the MSE for a single pixel. Then, PSNR with $m$-sampling is defined as follows:

$$\begin{aligned}&\mathrm{PSNR}_m = 10\log_{10}\left(\frac{255^2}{\mathrm{MSE}_m}\right) = 10\log_{10}(m) + C,\\ &\text{where } C = 10\log_{10}(\pi^2) \approx 9.94. \quad (21)\end{aligned}$$

This equation shows that PSNR increases linearly with the logarithm of $m$, independent of image content or size. In

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addition, the terms $p_i$ and $q_i$, which represent pixel intensities, have been removed, allowing PSNR to be estimated regardless of the image texture. Since a first-order approximation is applied, the higher-order terms $R_2$, which include expressions like $\frac{g(\hat{q}_i,q_i)}{m}$, retain image texture. As $m$ increases, these terms diminish, making higher-order information unnecessary for large trial counts.

### C. Theoretical formula for NEQR

Assuming there are no errors in setting the quantum bits, passing through the gates, or in the sampling itself, NEQR can perfectly reproduce a pixel with 8-bit accuracy at once. Therefore, we use the expected number of trials required for perfect restoration as an alternative metric, instead of PSNR. Note that this method does not depend on the image's entropy.

The expected number of trials for this problem is known as the *coupon collector's problem*. The expected number of samples, $E_N$, for which all $N$ pixels are sampled at least once is expressed below using the harmonic number $H_N$:

$$E_N = N \cdot \left(\frac{1}{1}+\frac{1}{2}+\cdots+\frac{1}{N}\right) = NH_N,\ H_N = \sum_{k=1}^{N}\frac{1}{k}. \quad (22)$$

## IV. Experiments and Results

We compared the similarity between ideal images and reconstructed quantum images for QL, FRQI, and NEQR with image sizes $2^n \times 2^n$ $(n = 1 \sim 13)$. Note that the evaluation of QL is the same as that of $1 \times 1$ FRQI. We performed these experiments on GPU-based simulations using cuQuantum.

Figure 1a is the original, and Figs. 1b, 1c, 1d are the restored FRQI images with 100K, 10M, and 1G samples, respectively. Figure 2 shows the reconstructed PSNR of FRQI for the number of samples per pixel. The theoretical PSNR is linear, and the measured PSNR is nearly linear. However, when the number of samples is small, the part is not on the line, since the value of the unsampled pixels is assumed to be 0.

Figure 3 shows PSNR calculated in two ways: the exact expansion of the binomial theorem (13) and the approximation (21). We assume that the image is uniform (i.e., $p_i$ is constant) in this case, and the pixel values range from 0 to 255. We limit $m = 2^{10}$ due to the numerical accuracy limit of the exact solution. The exact solution always samples 0 or 1 at both ends, corresponding to 0 and 255, resulting in a PSNR of $\infty$. The PSNR changes nonlinearly near both ends, whereas the approximation is a horizontal line with all pixel values equal. The approximation accurately represents the central area well. The accuracy near both ends decreases, but the degradation decreases as $m$ increases.

Table I shows the NEQR relationship between the simulated results and the theoretical formula. It demonstrates that the restoration is generally in accordance with the theory. It also shows that the number of samples required for restoration is significantly smaller than that of FRQI. The restored NEQR's 8-bit intensity is sampled by converging each bit to 0 or 1; thus, no analog errors occur.



(a) source (b) 100K (0.38/pix) (c) 10M (38.14/pix)(d) 1G (3814/pix)

Fig. 1: Original image and restorations from FRQI.

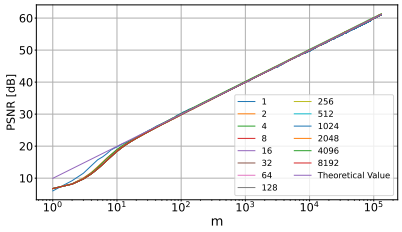



Fig. 2: FRQI: PSNR vs average samples per pixel (log scale).

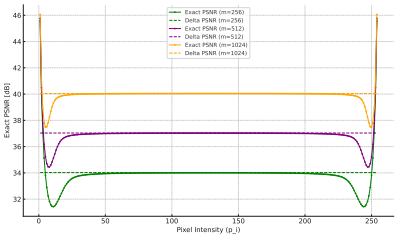


Fig. 3: The relationship between PSNR based on the exact solution formula and PSNR based on the approximate solution formula when FRQI $p_i \in [1:254]$ and $m$ are changed. Note that $p_i = 0, p_i = 255$ becomes PSNR$= \infty$.

TABLE I: Expected number of samples required for NEQR perfect restoration and expected and measured values for each pixel of the second order (average of 100 times)

| Size | ideal in total | ideal ($m$) | measure ($m$) |
|---|---|---|---|
| $2 \times 2$ | 8 | 2.000 | 2.084 |
| $4 \times 4$ | 54 | 3.375 | 3.383 |
| $8 \times 8$ | 303 | 4.734 | 4.790 |
| $16 \times 16$ | 1,567 | 6.121 | 6.226 |
| $32 \times 32$ | 7,689 | 7.509 | 7.425 |
| $64 \times 64$ | 36,451 | 8.899 | 8.866 |
| $128 \times 128$ | 167,532 | 10.227 | 10.359 |
| $256 \times 256$ | 754,970 | 11.520 | 11.647 |
| $512 \times 512$ | 3,361,228 | 12.821 | 12.887 |
| $1024 \times 1024$ | 14,737,887 | 14.056 | 14.445 |
| $2048 \times 2048$ | 63,830,302 | 15.220 | 15.769 |
| $4096 \times 4096$ | 274,870,634 | 16.387 | 16.963 |
| $8192 \times 8192$ | 1,172,155,032 | 17.462 | 17.828 |

## V. Conclusion

In this paper, we theoretically and numerically evaluated the upper bound quality for image restoration based on sampling for the three representative QIRs: QL, FRQI, and NEQR. The proposed theoretical framework suggests that reconstruction performance is solely determined by the average number of samples per pixel, regardless of image content and size. This prediction was successfully validated through GPU-based quantum simulations. Assuming error-free quantum computing, this analysis may underestimate the real challenges.

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